Diverse Text-Prompt Generation for Remote Sensing Image Classification

Wenda Zhao, *Member, IEEE*, Xiangzhu Lv, Ruikun He, Fan Zhao, Haipeng Wang, and You He

***Abstract*— Inadequate remote sensing image training data usually makes remote sensing image classification models achieve low accuracy. Thus, we propose a diverse text-prompt generation learning (DPL) method. The context optimization (CoOp) model transfers the feature extraction capabilities of the CLIP model to downstream tasks with learnable text prompts. However, the small number of samples in remote sensing images can easily lead to noise. In contrast, DPL introduces a diverse text-prompt generation structure. Due to the diversity of multiple prompts, noise generated by inadequate samples is suppressed. Moreover, in order to keep the prompts diverse, we propose a prompt diversity loss. This loss pulls the prompts away from each other, which suppresses the noise generated by the limited training samples. Extensive experiments show that our method achieves superior performance than the existing methods on DOTA, HRRSD, and NWPU VHR-10 datasets. The model and code are available at https://github.com/LvXiangzhu/DPL.**

***Index Terms*— Diverse text-prompt generation, prompt diversity loss, remote sensing image classification.**

I. Introduction

REMOTE sensing image classification [1], [2], [3] is a notable study subject nowadays. In reality, since collecting remote sensing samples needs high professionalism and the time cost of labeling, the labeled remote sensing samples are sparse. Thus, inadequate remote sensing samples are used to train the classification model, which tends to decrease the accuracy. For example, as shown in Fig. 1, trustworthy long-tailed classification (TLC) [4] trained on HRRSD training dataset performs only 84.1% accuracy on HRRSD testing dataset. Moreover, its generalization performance is only 82.1% and 27.5% on the NWPU VHR-10 dataset and DOTA dataset, respectively. Therefore, how to improve the accuracy of the model under the condition of remote sensing small samples is an important issue.

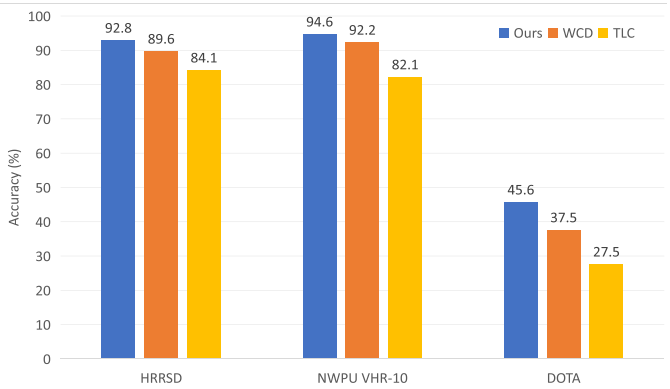


Fig. 1. Results of ours (DPL) and WCD [5], TLC [4] methods on HRRSD [6], NWPU VHR-10 [7], and DOTA [8] datasets. HRRSD dataset is used to train these models, and HRRSD, NWPU VHR-10, and DOTA datasets are used to test. Compared with the previous SOTA methods, our method achieves the highest accuracy on three datasets.

There are three kinds of methods to solve the small-sample image classification problem: meta-learning [9], transfer learning [10], and data augmentation [11]. Meta-learning is a method of learning "how to learn" through a double loop, thus converging to a point where one can quickly adapt to various tasks. However, the training of meta-learning is complex, so it is not suitable for remote sensing image datasets with ten to hundreds of samples per category. Transfer learning trains the baseline with a large number of samples and fine-tunes the pre-trained model with a small number of samples. However, in the case of small-sample remote sensing datasets, the model trained by transfer learning usually obtains noisy features, which decrease the accuracy of the model. Data augmentation increases the number of samples with traditional or deep learning methods. However, the quality of the enhanced samples is uneven, which will impact the accuracy of model.

CLIP model [12] is the first multimodal model that introduces a large language model into the image classification task. Because of the abundant datasets and the large-scale model, the CLIP model has strong feature extraction and generalization capabilities. However, the CLIP model cannot be adapted to the dataset in the field of remote sensing. Considering that the CLIP model has a large number of parameters, therefore, it is not practical to fine-tune CLIP to fit a professional dataset. Addressing this, context optimization (CoOp) model [13] fixes the text encoder and image encoder of CLIP and introduces learnable prompt parameters on the text

Received 24 June 2024; revised 29 September 2024 and 8 December 2024; accepted 16 December 2024. Date of publication 25 December 2024; date of current version 6 January 2025. This work was supported by the National Natural Science Foundation of China under Grant 62176038 and Grant 62001450. *(Corresponding author: Haipeng Wang.)*

Wenda Zhao, Xiangzhu Lv, and Ruikun He are with the Key Laboratory of Intelligent Control and Optimization for Industrial Equipment of Ministry of Education and the School of Information and Communication Engineering, Dalian University of Technology, Dalian 116024, China (e-mail: zhaowenda@dlut.edu.cn; lvxiangzhu@mail.dlut.edu.cn; heruikun@mail.dlut.edu.cn).

Fan Zhao is with the School of Physics and Electronic Technology, Liaoning Normal University, Dalian 116029, China (e-mail: Fan_Zhao20@163.com)

Haipeng Wang and You He are with the Research Institute of Information Fusion, Naval Aviation University, Yantai 264001, China (e-mail: whp5691@163.com; heyou_f@126.com).

Digital Object Identifier 10.1109/TGRS.2024.3522283

1558-0644 © 2024 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:53:07 UTC from IEEE Xplore. Restrictions apply.

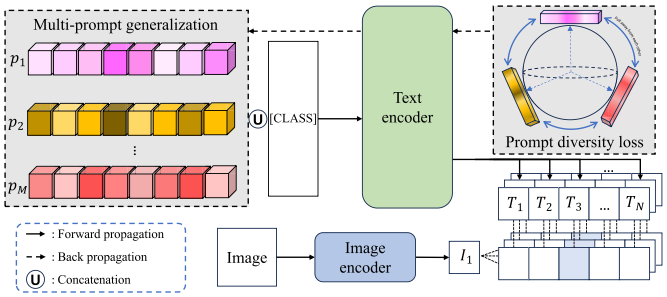


Fig. 2. Motivation of the proposed DPL method. In order to avoid prompt noise generated by a few-sample remote sensing image, multiprompt generation structure is designed. Meanwhile, prompt diversity loss is proposed to keep the diversity of multiple prompts.

side to fine-tune the model, thereby transferring the capabilities of the CLIP model to the remote sensing datasets. However, the CoOp model only has one learnable prompt, which is prone to generating noise, thus reducing the performance of the model. Therefore, we propose a diverse text-prompt generation learning (DPL) method to solve the problem of small-sample remote sensing image classification. First, we introduce a multiprompt generation structure on the text input side to make prompts diverse. On account of large differences in the location distribution of diversity prompts noise, the noise can be suppressed, thus improving the accuracy of text description. Second, we use prompt diversity loss to further constrain multiple prompts. By pulling away from each other in the feature space, the diversity of prompts is maintained, enhancing the model's performance.

Specifically, as shown in Fig. 2, we use multiple learnable prompts $\{p_1, p_2, \ldots, p_M\}$ on the text input side. Prompts with different categories form multiple sets of text features $\{t_1, t_2, \ldots, t_M\} = \{p_1 + [\text{CLASS}], p_2 + [\text{CLASS}], \ldots, p_M + [\text{CLASS}]\}$. However, if these prompts are not constrained, the model will learn the same prompts, which degenerates into a single-prompt model (CoOp). Therefore, in order to make the descriptive information of the prompt diverse, we propose a prompt diversity loss to constrain the distance of the prompts in the latent feature space. The prompt diversity loss makes multiple prompt features orthogonal in pairs, which pulls these prompt features far away from each other to the greatest extent. In addition, in order to ensure that diverse prompts do not deviate from original categories, we shorten the distance between multiple text prompt features and image features from the same category. Thus, the prompts have diverse features and yet are able to remain vicinity of the original categories.

The main contributions of this article are summarized in three folds.

1) This article studies remote sensing image classification with limited samples and builds the DPL method.
2) We design a multiprompt generation structure to suppress noise generated by insufficient samples and utilize prompt diversity loss to keep multiple prompts diverse to improve classification accuracy.
3) HRRSD, DOTA, and NWPU VHR-10 datasets are adopted for experiments, which exhibit that our method has superior performance.

II. Related Work

A. Image Classification

Image classification is an important and fundamental research task that aims to identify the category of the object in the image. For example, Li et al. [4] propose a TLC method to jointly conduct classification and uncertainty estimation to identify hard samples. Du et al. [14] propose a novel progressive multigranularity training framework to tackle fine-grained visual classification. Nam et al. [15] propose style-agnostic network to reduce the intrinsic style bias of CNNs to close the gap between domains. There are image classification models such as [16], which consider image classification problems from a variety of different perspectives. For example, Dosovitskiy et al. [16] discover that the Transformer model in natural language processing (NLP) can also be used to solve classification tasks in the visual domain and propose Vision Transformer (ViT). ViT model inputs several patches into Transformer structure as sequences and considers the class token as the final classification output. Further improvements to ViT have been made by methods such as [17]. Many image classification methods have also been proposed in the field of remote sensing image classification. For example, Zhao et al. [5] propose a weakly correlated distillation method to solve limited sample classification. Chen et al. [18] introduce RSMamba, a method based on the state-space model for remote sensing image classification. Zhao et al. [19] propose a center-wise feature consistency learning mechanism for long-tailed remote sensing object recognition. Li et al. [20] propose a Mamba classification model, which integrates spatial and spectral information in an adaptive manner that can simultaneously model long-range interaction of the whole image.

However, existing methods often adopt lots of samples to train, resulting in degraded performance on small-sample datasets. Therefore, we propose a DPL method, which uses fewer training samples to obtain superior classification accuracy.

B. Multimodal Learning

CLIP model is an important work in the field of image–text modality. Benefiting from large-scale training datasets and image–text feature alignment, the large-scale model has strong feature extraction and generalization abilities. This leads to a large number of related multimodal studies based on CLIP model. From a horizontal point of view, CLIP model is applied to many other tasks, such as semantic segmentation [21], object detection [22], and video domains [23]. From a longitudinal point of view, there are also many few-shot classification methods that are similar to CLIP model [24]. These methods can be summarized as prompt-based methods [13], adapter-based methods [25], and external knowledge-based methods [26]. For example, fixing the text encoder and visual encoder of CLIP, CoOp model [13] introduces a learnable text prompt vector and improves the performance of the model by training the prompt vector.

CLIP model is trained on a large number of natural image datasets, causing low accuracy performance on remote sensing

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:53:07 UTC from IEEE Xplore. Restrictions apply.

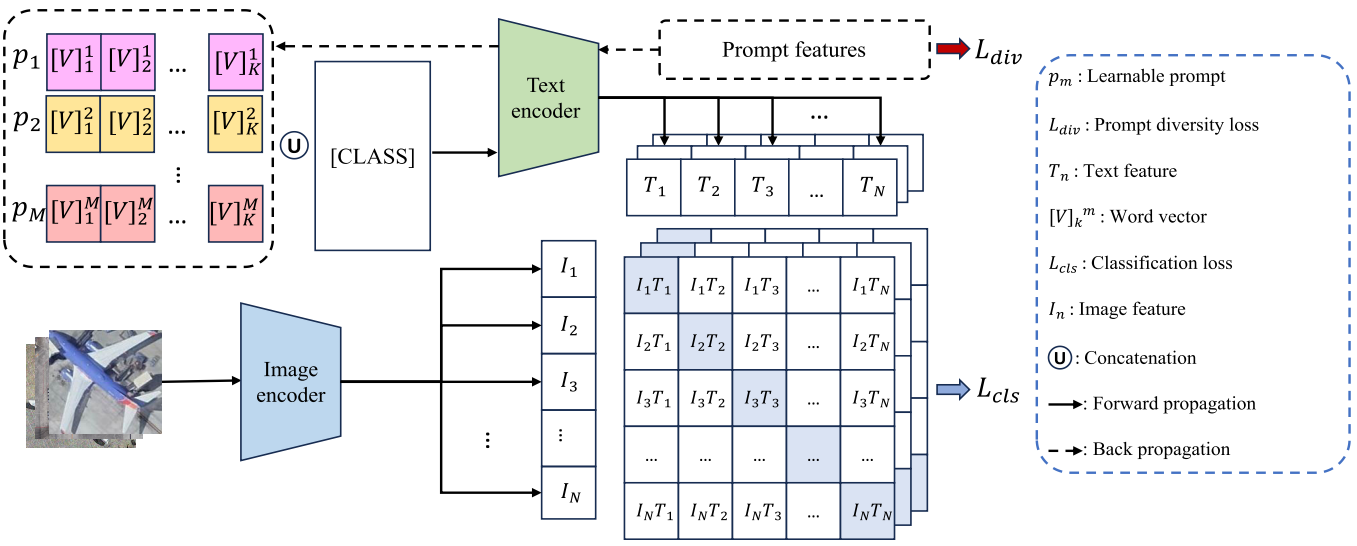


Fig. 3. Illustration of DPL. The multiprompt generation structure on the text side is used to suppress noise and increase the diversity of the model. In order to maintain the diversity of prompt features to make the prompt noise position distribute differently, the prompt diversity loss L_{div} is proposed to pull prompt features away from each other.

images. CoOp model can transfer the feature extraction capability of CLIP to the professional dataset with a learnable text prompt. However, training a model with a small number of samples will make noise and reduce the performance of the model. Therefore, we introduce diversity learning into the multimodal model to suppress the noise of the model through diverse prompts.

C. Diversity Learning

Diversity learning is the integration of multiple different models which have low accuracy, often achieving better performance than a single model [27]. Deep learning-based diversity learning methods include negative correlation learning [28], explicit/implicit ensemble method [29], and decision fusion strategies [30]. For example, Zhao et al. [28] propose a novel learning strategy by breaking DBD problem into multiple smaller defocus blur detectors. Thus, estimate errors can cancel out each other. Shi et al. [29] propose an ensemble deep Random Vector Functional Link network (edRVFL), which is obtained by training a single dRVFL network once. Zhao et al. [30] propose adaptive ensemble network (AENet) and encoder-feature ensemble network (EFENet) to focus on boosting diversity while costing less computation. Inspired by those works, we introduce diverse text-prompt generation into multimodal large model to achieve superior accuracy on small-sample remote sensing datasets.

III. Diverse Text-Prompt Generation

A. Motivation and Framework

In recent years, multimodal classification has provided a direction for improving the accuracy of small-sample remote sensing images. CLIP model [12] is the first multimodal model that has excellent performance by combining large-scale models in NLP domain with the image classification task. However, CLIP model trained with natural images cannot perform well in the specialized remote sensing image classification field. In addition, because of the large scale of

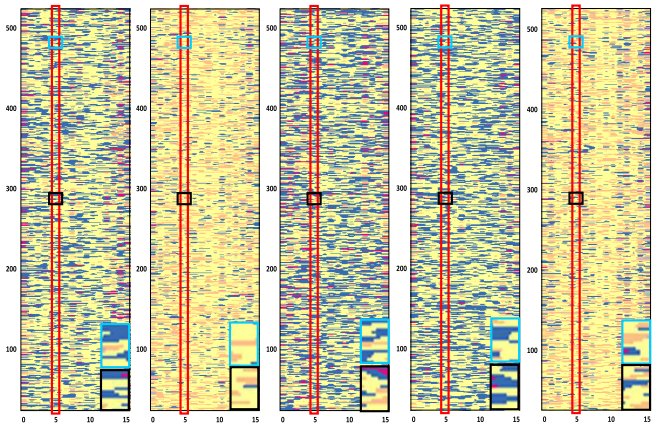


Fig. 4. Visualization of the multiprompt generation structure. The abscissa is the number of word vectors. The ordinate is the dimension of a word vector.

CLIP model, naively fine-tuning CLIP model can easily lead to model overfitting. Zhou et al. [13] find that different text prompts have a great impact on the classification accuracy of CLIP model. Therefore, in order to solve the above problem, CoOp model fixes the image encoder and text encoder of CLIP and adds a learnable text prompt parameters, so that the model automatically learns a text prompt that is suitable for the current dataset. However, for small-sample remote sensing datasets, the prompt parameters learned by the single-prompt model are prone to make error. Therefore, we propose a DPL method. As shown in Fig. 3, the proposed method aims to offset the descriptive error in the prompts through diversity learning, thus improving the performance of the model.

B. Multiprompt Generation

CoOp model improves the accuracy of the model by learning a suitable prompt. However, when trained on a small-sample remote sensing dataset, a single text prompt can easily generate noise, resulting in descriptive error, which reduces the classification performance of the model.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:53:07 UTC from IEEE Xplore. Restrictions apply.

The reason may be explained as follows. Given many prompts $(p_m, m = 1, \ldots, M)$, we integrate them by ensemble learning and then obtain an average prompt $\hat{p}$

$$\hat{p} = \frac{1}{M}\sum_{m=1}^{M} p_i. \tag{1}$$

Typically, for an average prompt $\hat{p}$ with the ground truth Y, the expectation generalization error can be calculated by the bias-variance decomposition algorithm [28], [31] as

$$E(\hat{p} - Y)^2 = (E\{\hat{p}\} - Y)^2 + E(\hat{p} - E\{\hat{p}\})^2 \tag{2}$$

where E stands for an expectation operator. Then, we can obtain

$$E\{(\hat{p} - Y)^2\} = \frac{1}{M}\left(\sum_{m=1}^{M} E\{(p_m - Y)^2\} - \frac{1}{M}\sum_{m=1}^{M} E\{(p_m - \hat{p})^2\}\right) \tag{3}$$

where the first item indicates the noise between a single prompt $p_m, m = 1, \ldots, M$ and the ground truth Y, and the second item represents the diversity of the generated prompts. When trained with a small sample size, the model generalization decreases which will result in a larger value of $E\{(\hat{p} - Y)^2\}$. Additionally, diverse prompts will lead to a higher value of the second item. Therefore, the first item becomes larger, i.e., noise is introduced.

In order to avoid the above drawback, we introduce multiprompt on the text side. The parameters of the prompts are not shared, so different prompts can be learned. The descriptive errors of different prompts are located differently, thus can be canceled out. Specifically, the formula for multiprompt generation structure is $\{p_1, p_2, \ldots, p_M\}$, where $p_m = [V]_1^m[V]_2^m, \ldots, [V]_K^m, (m \in \{1, 2, \ldots, M\})$ is the basic structure of a prompt, $[V]_K^m, (k \in \{1, 2, \ldots, K\})$ is a word vector which has the same dimension of 512 with other word vectors, and K represents the number of word vectors. A text vector consists of a text prompt and a class, as shown in the formula

$$t_m = p_m + [\text{CLASS}] = [V]_1^m[V]_2^m, \ldots, [V]_K^m[\text{CLASS}]. \tag{4}$$

Multiple prompts make up the text vector

$$\{t_1, t_2, \ldots, t_M\} = \{p_1 + [\text{CLASS}], p_2 + [\text{CLASS}], \ldots, p_M + [\text{CLASS}]\}. \tag{5}$$

All categories share every prompt. We visualize multiple prompts, as shown in Fig. 4. Obviously, five prompts are different, e.g., the prompts in the blue box and black box, indicating the diversity of multiple prompts.

C. Prompt Diversity Loss

In order to avoid the prompts learning the same parameters, we propose a prompt diversity loss as a constraint for the prompts, as shown in Fig. 5. Specifically, the prompts $\{p_1, p_2, \ldots, p_M\}$ are input into the text encoder to obtain the prompt feature $\{f_1, f_2, \ldots, f_M\}$. Then, the prompt diversity loss is used to pull the prompt features away from each other to increase the diversity of the prompts, as shown in the formula

$$L_{\text{div}} = \sum_{i=1}^{M}\sum_{j=i+1}^{M} \text{sim}(f_i, f_j), \quad i \neq j \tag{6}$$

where f_i and f_j are two different prompt features, $\text{sim}(\cdot)$ is the cosine similarity, i.e., $\text{sim}(x, y) = ((x \cdot y)/(|x||y|))$, and M stands for the number of prompts. We provide a simple explanation of this method as follows. The prompt features mapped to the latent space contain noise, which will interfere with the classification effect. Therefore, in order to keep the prompts diverse, the prompt diversity loss is used to pull the prompts away from each other. To be specific, the cosine similarity is used to keep the prompt features orthogonal in pairs, thus keeping the difference in the noise distribution of different prompts as large as possible, thereby canceling out the noise. M prompts $\{f_1, f_2, \ldots, f_M\}$ have different noise distributions. When the noise of one feature is superimposed with multiple noise-free features at that location, the noise is diluted, resulting in a virtually noise-free feature.

In order to integrate the prompt features and avoid the classification inaccuracy caused by too many prompt features, we use the prompt integration loss as a constraint, and the formula is as follows:

$$L_{\text{cls}} = \frac{1}{M}\sum_{m=1}^{M} \frac{\exp\left(\text{sim}\left(f_I, f_{Tm}^{+}\right)/\tau\right)}{\sum_{j=1}^{C}\exp\left(\text{sim}\left(f_I, f_{Tm}^{j}\right)/\tau\right)} \tag{7}$$

where f_I denotes the image feature, f_T represents the text feature, f_{Tm}^{+} signifies the text feature that is the same category as the image feature f_I, τ is the temperature hyperparameter, $\text{sim}(\cdot)$ stands for the cosine similarity, C means the number of categories, and M indicates the number of prompts. As shown in Fig. 6, since multiple prompts are restrained by the diverse orthogonal loss, their noise distributions are different. Therefore, the noise is suppressed when the multiple prompts are integrated, i.e., the descriptive error is offset.

Thus, the total loss of the network is shown as follows:

$$L = L_{\text{cls}} + \beta L_{\text{div}} \tag{8}$$

where β is a hyperparameter.

In summary, the primary innovation includes two aspects: 1) the prompt diversity loss is designed to generate diverse prompts with different noise distributions and 2) the prompt integration loss is introduced to suppress descriptive errors that improve the classification accuracy.

D. Optimization and Inference

1) Optimization: The optimization process is shown in Algorithm 1. Considering the effectiveness, the number of prompts M is taken to 5. ViT-32 network [16] is adopted as the image encoder, and the transformer [32] is selected as the text encoder. Notably, both encoders are fixed. During optimization, the prompt vectors $\{p_1, p_2, \ldots, p_M\}$ are fed into the text encoder, and then the prompt parameters are updated with prompt diversity loss. Text vectors

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:53:07 UTC from IEEE Xplore. Restrictions apply.

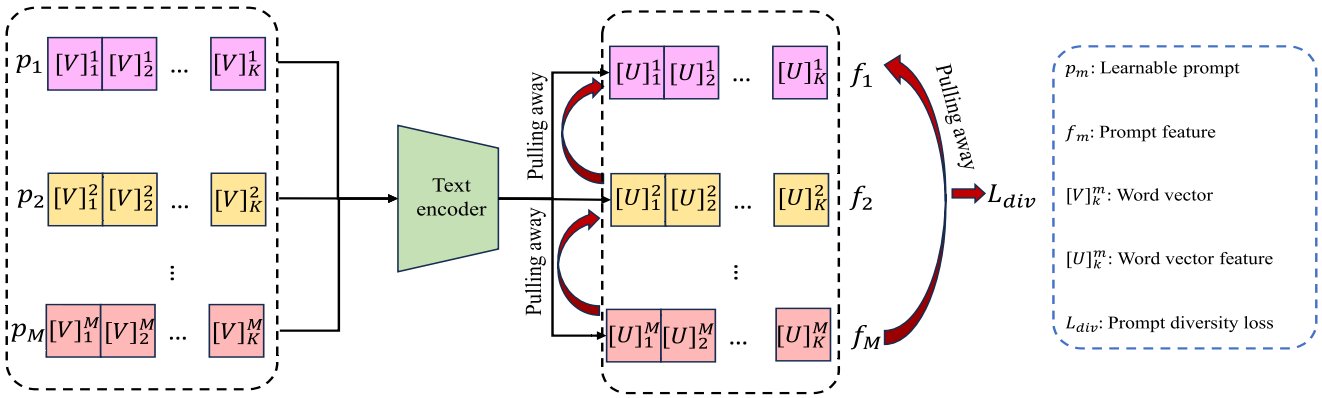


Fig. 5. Illustration of prompt diversity loss. Multiple prompt vectors are input into the text encoder to obtain prompt features. Prompt features are pulled away from each other by the prompt diversity loss.

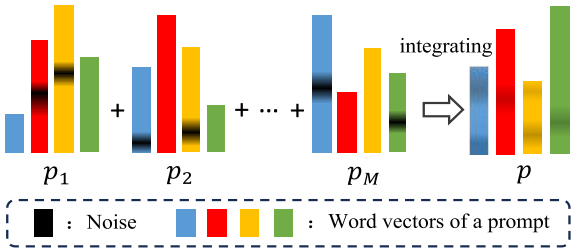


Fig. 6. Illustration of the noise suppression. Prompts $\{p_1, p_2, \ldots, p_M\}$ contain different noise distributions. Thus, the noise is suppressed when the prompts are aggregated.

Algorithm 1 Optimization and Inference Process of Our Method

Optimization:

1. Fix the image encoder and the text encoder.
2. Input the remote-sensing dataset to the image encoder.
3. Input the text vectors to the text encoder.
4. Optimize the prompts by Equation (6) and Equation (7).

Inference:

1. Input a remote-sensing image to the image encoder.
2. Input the trained text vectors to the text encoder.
3. Calculate the similarity of the image-text pair to obtain the class probability.

$\{t_1, t_2, \ldots, t_M\}$ and images are input into the text encoder and image encoder, respectively. Then, multiple text features are integrated with classification loss to update the prompt parameters.

2) Inference: As shown in Algorithm 1, first, multiple text vectors $\{t_1, t_2, \ldots, t_M\}$ are fed into the text encoder to generate multiple text features $\{f_{T1}^1, f_{T1}^2, \ldots, f_{T1}^C, \ldots, f_{TM}^C\}$, where C is the number of categories and M represents the number of prompts. We input an image to the image encoder that obtains an image feature f_I. Second, the similarity between each set of text features and the image feature is calculated to form a probability vector. Then, the probability vector is averaged to obtain the final result. The category with the highest probability is the corresponding category, as shown in (7).

IV. Experiment

A. Configuration

1) Dataset: We use three datasets: HRRSD dataset [6], NWPU VHR-10 dataset [7], and DOTA dataset [6]. Our DPL model is trained with HRRSD dataset and tested on HRRSD dataset to verify the accuracy of the model. Furthermore, the model is tested on NWPU VHR-10 dataset and DOTA dataset to illustrate the generalization of the model. Table I describes the category distribution of three datasets. We select ten common categories of the three datasets as the experimental dataset.

2) Implementation: The graphics card used in the experiment is a GTX 2080Ti GPU. The deep learning framework we use is PyTorch. The optimizer is SDG, and the learning rate is taken to 10^{-5}. The momentum and weight decay are taken to 0.9 and 0.0005, respectively. The training batch size and the test batch size are set to 32 and 100, respectively. Training epochs are 1000. In order to unify the feature dimensions of the fully connected layer, the input images are resized to 224 × 224 pixels. The number of prompts is 5 and the hyperparameter β is 5. The temperature hyperparameter τ is 1. Accuracy (Acc.) metric is used to validate the classification performance.

B. Hyperparameter Settings

1) Effectiveness of Prompt Diversity Loss: In order to make the prompts have abundant diversity, we propose prompt diversity loss. This loss makes the prompts orthogonal to each other, pulling the prompts away from each other to the greatest extent. In this way, the prompt noise caused by insufficient samples is canceled, and the recognition accuracy of the model is improved. In this experiment, three prompts are used as examples to test the results of prompt diversity loss with different weights β on the HRRSD dataset. Then, generalization is tested on the NWPU VHR-10 dataset and DOTA dataset, as shown in Table II. $\beta = 0$ indicates that there is no prompt diversity loss and only a classification loss cannot offset the noise. Thus, the performance of the model is limited. When the weight $\beta < 5$, the performance of the model gets better as β increases. This is because the prompt diversity loss makes the model generate diverse prompts.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:53:07 UTC from IEEE Xplore. Restrictions apply.

TABLE I
TRAINING AND TESTING SAMPLE STATISTICS

Name	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9
Category	Baseball-diamond	Basketball-court	Bridge	Ground-track-field	Harbor	Airplane	Ship	Vehicle	Storage-tank	Tennis-court
HRRSD-Train	101	99	116	96	98	124	97	119	110	109
HRRSD-Test	1001	832	1090	795	938	1217	935	1178	1095	1011
DOTA-Test	214	132	464	144	2090	2531	8960	5438	2888	760
NWPU VHR-10-Test	390	159	124	163	224	757	302	598	655	524

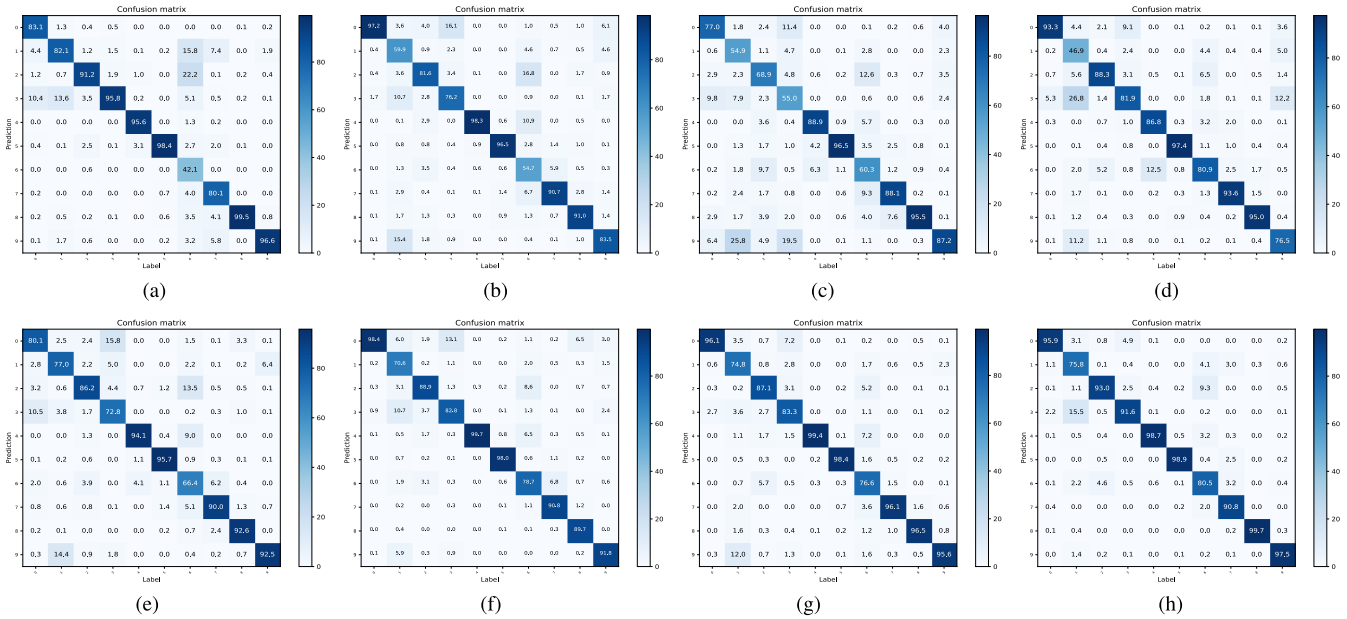


Fig. 7. Confusion matrices are generated from the results of different methods using HRRSD. (a) PMG. (b) TLC. (c) SagNet. (d) RIDE. (e) CWFC. (f) WCD. (g) RSMamba. (h) Ours.

TABLE II
EFFECTIVENESS AND GENERALIZATION STUDY OF PROMPT DIVERSITY LOSS. OUR DPL MODEL IS TRAINED USING HRRSD TRAINING DATA AND TESTED USING HRRSD, DOTA, AND NWPU VHR-10 DATASETS. OVERALL ACCURACY (%) IS USED FOR EVALUATION

β	HRRSD	NWPU VHR-10	DOTA
$\beta = 0$	91.9	93.4	42.8
$\beta = 1$	92.5	93.6	41.9
$\beta = 5$	**92.8**	94.6	**45.6**
$\beta = 10$	92.3	**94.9**	45.0

TABLE III
EFFECTIVENESS AND GENERALIZATION STUDY OF MULTIPROMPT. OUR DPL MODEL IS TRAINED USING HRRSD TRAINING DATA AND TESTED USING HRRSD, DOTA, AND NWPU VHR-10 DATASETS. OVERALL ACCURACY (%) IS USED FOR EVALUATION

M	HRRSD	NWPU VHR-10	DOTA
$M = 1$	91.6	92.5	39.4
$M = 3$	92.5	93.5	43.3
$M = 5$	**92.8**	**94.6**	**45.6**
$M = 7$	92.6	94.2	45.0
$M = 9$	92.4	93.5	43.7

When $\beta > 5$, the performance of the model decreases on the HRRSD dataset and does not have a significant change on the DOTA dataset and NWPU VHR-10 dataset as β increases. Since the proportion of prompt diversity loss is large enough to generate diverse prompts that offset noise. Instead, too much prompt diversity loss makes it easy to impact classification loss. When $\beta = 5$, the model has the best performance of 92.8% on the HRRSD dataset and the best generalization of 45.6% on the DOTA dataset. Therefore, β is set to 5.

2) Effectiveness of Multiprompt: A single prompt can only describe a single piece of textual information. Therefore, generated noise limits the performance of the model. Solving this, we adopt the model structure of multiprompt generation. The noise of multiple prompts at different positions cancels each other out, thereby improving the performance of the model. In order to verify the effectiveness of multiprompt, several different prompt branches are used. The number of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:53:07 UTC from IEEE Xplore. Restrictions apply.

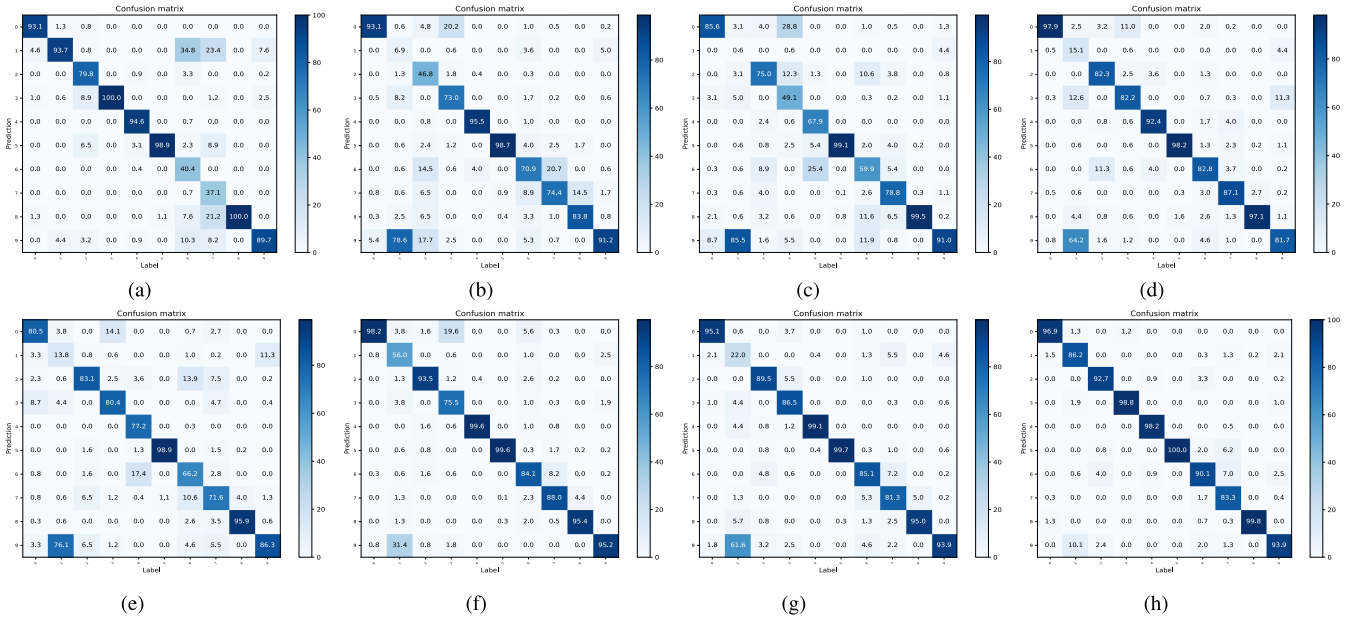


Fig. 8. Confusion matrices are generated from the results of different methods using NWPU VHR-10. (a) PMG. (b) TLC. (c) SagNet. (d) RIDE. (e) CWFC. (f) WCD. (g) RSMamba. (h) Ours.

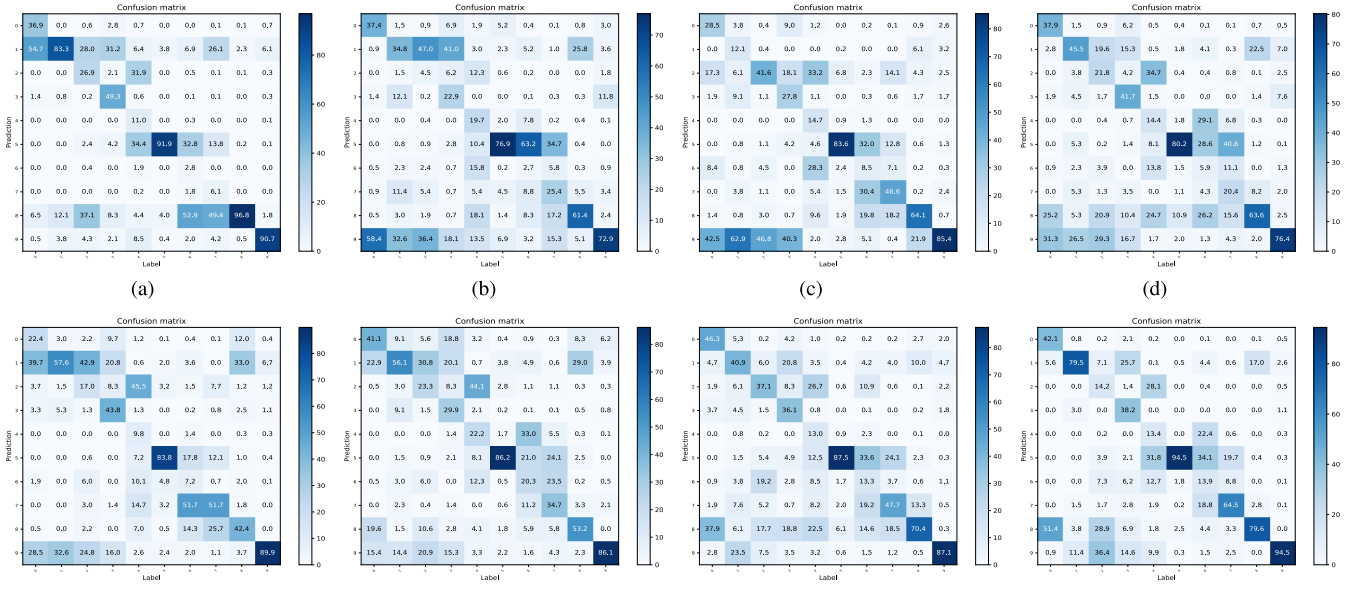


Fig. 9. Confusion matrices are generated from the results of different methods using DOTA. (a) PMG. (b) TLC. (c) SagNet. (d) RIDE. (e) CWFC. (f) WCD. (g) RSMamba. (h) Ours.

TABLE IV

INFLUENCE OF DIFFERENT NUMBERS OF PROMPT WORD VECTORS K. OUR DPL MODEL IS TRAINED USING HRRSD TRAINING DATASET AND TESTED USING HRRSD, DOTA, AND NWPU VHR-10 DATASETS, WHERE EVALUATION METRIC IS OVERALL ACCURACY (%)

K	HRRSD	NWPU VHR-10	DOTA
K = 8	92.5	93.6	41.9
K = 16	**92.8**	**94.6**	**45.6**
K = 24	92.6	93.8	43.2

TABLE V

MODEL IS TRAINED ON DIFFERENT NUMBERS OF TRAINING SAMPLES OF HRRSD DATASET AND IS TESTED USING HRRSD, NWPU VHR-10, AND DOTA DATASETS. OVERALL ACCURACY (%) IS USED FOR EVALUATION

Setting	HRRSD	NWPU VHR-10	DOTA
HRRSD	92.8	94.6	45.6
HRRSD-1/2	92.1	93.9	46.5
HRRSD-1/4	90.6	93.8	43.3
HRRSD-1/8	88.1	89.0	39.7

prompts M is set to 1, 3, 5, 7, and 9, respectively, where the model has one prompt, which turns it into CoOp model, and the rest are multiprompt models. We verify the accuracy on the HRRSD dataset and generalization on NWPU VHR-10 dataset and DOTA dataset, respectively, as shown in Table III. When the prompt number M is less than 5, the performance of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:53:07 UTC from IEEE Xplore. Restrictions apply.

TABLE VI

QUANTITATIVE COMPARISON OF DIFFERENT METHODS USING HRRSD DATASET. THE OVERALL ACCURACY (ACC.)(%) AND EACH CLASS ACCURACY ARE LISTED

Method	Acc.	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9
PMG	86.9	83.1	**82.1**	91.2	**95.8**	95.6	98.4	42.1	80.1	99.5	96.6
TLC	84.1	97.2	59.9	81.6	76.2	98.3	96.5	54.7	90.7	91.0	83.5
SagNet	78.8	77.0	54.9	68.9	55.0	88.9	96.5	60.3	88.1	95.5	87.2
RIDE	85.3	93.3	46.9	88.3	81.9	86.8	97.4	**80.9**	93.6	95.0	76.5
CWFC	85.6	80.1	77.0	86.2	72.8	94.1	95.7	66.4	90.0	92.6	92.5
WCD	89.6	**98.4**	70.6	88.9	82.8	**99.7**	98.0	78.7	90.8	89.7	91.8
RSMamba	90.4	96.1	74.8	87.1	83.3	99.4	98.4	76.6	**96.1**	96.5	95.6
Ours	**92.8**	95.9	75.8	**93.0**	91.6	98.7	**98.9**	80.5	90.8	**99.7**	**97.5**

TABLE VII

QUANTITATIVE COMPARISON OF DIFFERENT METHODS USING NWPU VHR-10 DATASET. THE OVERALL ACCURACY (ACC.)(%) AND EACH CLASS ACCURACY ARE LISTED

Method	Acc.	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9
PMG	82.2	93.1	**93.7**	79.8	**100.0**	94.6	98.9	40.4	37.1	**100.0**	89.7
TLC	82.1	93.1	6.9	46.8	73.0	95.5	98.7	70.9	74.4	83.8	91.2
SagNet	81.9	85.6	0.6	75.0	49.1	67.9	99.1	59.9	78.8	99.5	91.0
RIDE	88.0	97.9	15.1	82.3	82.2	92.4	98.2	82.8	87.1	97.1	81.7
CWFC	82.1	80.5	13.8	83.1	80.4	77.2	98.9	66.2	71.6	95.9	86.3
WCD	92.2	**98.2**	56.0	**93.5**	75.5	**99.6**	99.6	84.1	**88.0**	95.4	**95.2**
RSMamba	84.6	95.1	22.0	89.5	86.5	99.1	99.7	85.1	81.3	95.0	93.9
Ours	**94.6**	96.9	86.2	92.7	98.8	98.2	**100.0**	**90.1**	83.3	99.8	93.9

TABLE VIII

QUANTITATIVE COMPARISON OF DIFFERENT METHODS USING DOTA DATASET. THE OVERALL ACCURACY (ACC.)(%) AND EACH CLASS ACCURACY ARE LISTED

Method	Acc.	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9
PMG	29.7	36.9	**83.3**	26.9	**49.3**	11.0	91.9	2.8	6.1	**96.8**	90.7
TLC	27.5	37.4	34.8	4.5	22.9	19.7	76.9	2.7	25.4	61.4	72.9
SagNet	36.1	28.5	12.1	**41.6**	27.8	14.7	83.6	8.5	46.6	64.1	85.4
RIDE	28.3	37.9	45.5	21.8	41.7	14.4	80.2	5.9	20.4	63.6	76.4
CWFC	33.7	22.4	57.6	17.0	43.8	9.8	83.8	7.2	51.7	42.4	89.9
WCD	37.5	41.1	56.1	23.3	29.9	**22.2**	86.2	**20.3**	34.7	53.2	86.1
RSMamba	36.9	**46.3**	40.9	37.1	36.1	13.0	87.5	13.3	47.7	70.4	87.1
Ours	**45.6**	42.1	79.5	14.2	38.2	13.4	**94.5**	13.9	**64.5**	79.6	**94.5**

the model gradually increases as M increases. However, when M is greater than 5, the performance of the model decreases. Because when M is 5, the noise generated by prompts can be well offset. But as M continues to increase, the prompts will bring more noise that cannot be effectively offset, thereby reducing the performance of the model. The performance of the multiprompt model on HRRSD is better than that of the CoOp model. The performance of the five prompts on HRRSD is the best, with an accuracy of 92.8%. The generalization of multiple prompts is better than that of one prompt. Besides, the generalization of the five prompts on the two datasets is optimal, which is 94.6% on NWPU VHR-10 dataset and 45.6% on DOTA dataset. Thus, M is taken to 5.

3) Influence of Prompt Word Vectors: Here, we explore the effect of the prompts with different numbers of word vectors K. We set K to 8, 16, and 24, respectively. Then, we observe the performance of the model on the HRRSD dataset, NWPU VHR-10 dataset, and DOTA dataset, as shown in Table IV. When the word vector number $K < 16$, the performance of the model increases as K increases, since more word vectors can describe the image features completely. However, the performance of the model gets worse when $K > 16$, because too many word vectors will cause noise. When $K = 16$, the model has the best accuracy and generalization, which achieves 92.8%, 94.6%, and 45.6% on HRRSD, NWPU VHR-10, and DOTA datasets. Here, K is taken to 16.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:53:07 UTC from IEEE Xplore. Restrictions apply.

C. Effect of the Number of Training Samples

We study the effect of the number of training samples by decreasing HRRSD train samples by 1/2 (HRRSD-1/2), 1/4 (HRRSD-1/4), and 1/8 (HRRSD-1/8). The performance of the model trained on different numbers of training samples is shown in Table V. As can be seen when the number of training samples decreases, the performance of the model does not decrease much. This is because diverse prompts have the ability to suppress noise.

D. Comparison With State of the Arts

In this experiment, our DPL method is compared with traditional state-of-the-art (SOTA) methods, including PMG [14], SagNet [15], RSMamba [18], CWFC [19], WCD [5], and TLC [4], respectively. We observe the accuracy of these models on the HRRSD test dataset and the generalization on the NWPU VHR-10 test dataset and DOTA test dataset.

The accuracy of all methods on the HRRSD test dataset is shown in Table VI. Our DPL method achieves an accuracy of 92.8%, which is the best performance compared with traditional SOTA methods. Specifically, our method improves SagNet, TLC, RIDE, CWFC, PMG, WCD, and RSMamba by 14%, 8.7%, 7.5%, 7.2%, 5.9%, 3.2%, and 2.4%, respectively.

1) Generalization: We further validate the generalization of the diverse prompt method by comparing it with the previous SOTA methods on the NWPU VHR-10 dataset and DOTA dataset, as shown in Tables VII and VIII. As can be seen from these two tables, our DPL method achieves the best accuracy on both the NWPU VHR-10 dataset and the DOTA dataset. The results show that our method has the best generalization performance compared with other methods. Specifically, the accuracy of ours on the NWPU VHR-10 dataset is 94.6%, which is higher than the suboptimal method by 2.4% and higher than the average of all methods by 9.9%. The accuracy of ours on the DOTA dataset is 45.6%, which is higher than the suboptimal method by 8.1% and higher than the average of all methods by 12.8%.

There are several reasons why our DPL method has superior accuracy and generalization for small-sample remote sensing datasets: 1) we introduce multiprompt generation structure, where multiple different prompts have varied descriptive information of the text. Thus, the features can be represented adequately and 2) prompt diversity loss is used for pulling prompts away from each other, thereby minimizing incorrect descriptive information.

2) Visualization: We visualize the performance of different methods on the HRRSD dataset, NWPU VHR-10 dataset, and DOTA dataset, as shown in Figs. 7–9. The abscissa represents the true category, and the ordinate represents the predicted category. Each column represents the prediction result for the current category. Since some decimal parts are discarded, the sum of the elements is approximately equal to 1. As shown in Fig. 7, our DPL method achieves the best performance on the HRRSD dataset. As shown in Figs. 8 and 9, our DPL method has the best generalization performance on NWPU VHR-10 dataset and DOTA dataset.

V. Conclusion

We propose a DPL method for improving the accuracy and generalization of the model on small-sample remote sensing samples. First, we introduce the multiprompt generation structure that uses multiple different prompts to increase diversity. This avoids the noise description in a single prompt. Second, to ensure that the descriptive error distribution of multiple prompts is different, a prompt diversity loss is proposed. By pulling away the prompts from each other, the noise of the prompt features is canceled and the performance of the model is improved. Extensive experiments on three remote sensing datasets demonstrate the effectiveness of our method.

References

[1] W. Zhao, R. Yang, Y. Liu, and Y. He, "Style-content metric learning for multidomain remote sensing object recognition," in *Proc. AAAI Conf. Artif. Intell.*, vol. 37, no. 3, 2023, pp. 3624–3632.

[2] W. Zhao, J. Liu, Y. Liu, F. Zhao, Y. He, and H. Lu, "Teaching teachers first and then student: Hierarchical distillation to improve long-tailed object recognition in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5624412.

[3] W. Zhao, T. Tong, H. Wang, F. Zhao, Y. He, and H. Lu, "Diversity consistency learning for remote-sensing object recognition with limited labels," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5631510.

[4] B. Li, Z. Han, H. Li, H. Fu, and C. Zhang, "Trustworthy long-tailed classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 6970–6979.

[5] W. Zhao, X. Lv, H. Wang, Y. Liu, Y. He, and H. Lu, "Weakly correlated distillation for remote sensing object recognition," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5621011.

[6] Y. Zhang, Y. Yuan, Y. Feng, and X. Lu, "Hierarchical and robust convolutional neural network for very high-resolution remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5535–5548, Aug. 2019.

[7] G. Cheng, P. Zhou, and J. Han, "Learning rotation-invariant convolutional neural networks for object detection in VHR optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7405–7415, Dec. 2016.

[8] G.-S. Xia et al., "DOTA: A large-scale dataset for object detection in aerial images," *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jan. 2017, pp. 3974–3983.

[9] S. X. Hu et al., "Pushing the limits of simple pipelines for few-shot learning: External data and fine-tuning make a difference," *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, pp. 9058–9067, Jun. 2022.

[10] J. Xiao, L. Li, C. Wang, Z.-J. Zha, and Q. Huang, "Few shot generative model adaption via relaxed spatial structural alignment," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 11204–11213.

[11] K. Zhang, Z. Cao, and J. Wu, "Circular shift: An effective data augmentation method for convolutional neural network on image classification," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2020, pp. 1676–1680.

[12] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. ICML*, 2021, pp. 8748–8763.

[13] K. Zhou, J. Yang, C. C. Loy, and Z. Liu, "Learning to prompt for vision-language models," *Int. J. Comput. Vis.*, vol. 130, no. 9, pp. 2337–2348, Sep. 2022.

[14] R. Du et al., "Fine-grained visual classification via progressive multi-granularity training of jigsaw patches," in *Proc. Eur. Conf. Comput. Vis.* Berlin, Germany: Springer, 2020, pp. 153–168.

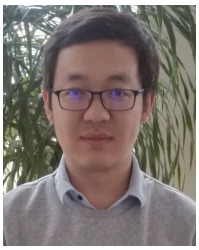
[15] H. Nam, H. Lee, J. Park, W. Yoon, and D. Yoo, "Reducing domain gap by reducing style bias," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 8690–8699.

[16] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[17] L. Zhu, X. Wang, Z. Ke, W. Zhang, and R. W. Lau, "BiFormer: Vision transformer with bi-level routing attention," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 10323–10333.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:53:07 UTC from IEEE Xplore. Restrictions apply.

[18] K. Chen, B. Chen, C. Liu, W. Li, Z. Zou, and Z. Shi, "RSMamba: Remote sensing image classification with state space model," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.
[19] W. Zhao, Z. Zhang, J. Liu, Y. Liu, Y. He, and H. Lu, "Center-wise feature consistency learning for long-tailed remote sensing object recognition," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5620011.
[20] Y. Li, Y. Luo, L. Zhang, Z. Wang, and B. Du, "MambaHSI: Spatial–spectral mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5524216.
[21] B. Li, K. Q. Weinberger, S. Belongie, V. Koltun, and R. Ranftl, "Language-driven semantic segmentation," 2022, *arXiv:2201.03546*.
[22] L. H. Li et al., "Grounded language-image pre-training," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 10955–10965.
[23] M. Wang, J. Xing, and Y. Liu, "ActionCLIP: A new paradigm for video action recognition," 2021, *arXiv:2109.08472*.
[24] F. Liu, T. Zhang, W. Dai, W. Cai, X. Zhou, and D. Chen, "Few-shot adaptation of multi-modal foundation models: A survey," 2024, *arXiv:2401.01736*.
[25] G. Peng et al., "Better vision-language models with feature adapters," 2021, *arXiv:2110.04544*.
[26] T. Ge et al., "Extensible prompts for language models on zero-shot language style customization," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2022, pp. 35576–35591.
[27] M. A. Ganaie, M. Hu, A. Malik, M. Tanveer, and P. Suganthan, "Ensemble deep learning: A review," *Eng. Appl. Artif. Intell.*, vol. 115, Sep. 2022, p. 105151.
[28] W. Zhao, B. Zheng, Q. Lin, and H. Lu, "Enhancing diversity of defocus blur detectors via cross-ensemble network," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 8897–8905.
[29] Q. Shi, R. Katuwal, P. N. Suganthan, and M. Tanveer, "Random vector functional link neural network based ensemble deep learning," *Pattern Recognit.*, vol. 117, Sep. 2021, Art. no. 107978.
[30] W. Zhao, X. Hou, Y. He, and H. Lu, "Defocus blur detection via boosting diversity of deep ensemble networks," *IEEE Trans. Image Process.*, vol. 30, pp. 5426–5438, 2021.
[31] G. Brown, J. L. Wyatt, P. Tino, and Y. Bengio, "Managing diversity in regression ensembles," *J. Mach. Learn. Res.*, vol. 6, no. 55, pp. 1621–1650, Dec. 2005.
[32] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, Jun. 2017, pp. 5998–6008.



Wenda Zhao (Member, IEEE) received the B.S. degree from Jilin University, Changchun, China, in 2011, and the M.S. and Ph.D. degrees from the University of Chinese Academy of Sciences, Beijing, China, in 2016.

He now is an Associate Professor with the School of Information and Communication Engineering, Dalian University of Technology (DUT), Dalian, China. His research interests include deep learning and computer vision tasks.



Xiangzhu Lv received the B.S. and M.S. degrees from Dalian University of Technology, Dalian, China, in 2021 and 2024, respectively.

Her research interests include object classification.



Ruikun He received the B.S. degree from Dalian University of Technology, Dalian, China, in 2023, where he is currently pursuing the M.S. degree.

His research interests include object classification and domain generalization.



Fan Zhao received the B.S. degree from Changchun University of Science and Technology, Changchun, China, in 2011, and the M.S. and Ph.D. degrees from Changchun Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences, Changchun, in 2014 and 2017, respectively.

She now is a Faculty Member with the School of Physics and Electronic Technology, Liaoning Normal University, Dalian, China. Her research interests include low-level vision tasks.



Haipeng Wang received the Ph.D. degree from Naval Aviation University, Yantai, China, in 2012.

He is currently a Professor with Naval Aviation University. His research interests include the general area of intelligent perception and fusion, and big data technology and application.



You He received the Ph.D. degree from Tsinghua University, Beijing, China, in 1997.

He is currently a Professor with the Department of Electronic Engineering, Tsinghua University, and the Director of the Institute of Information Fusion, Naval Aeronautical University, Yantai, China. His main research interests include information fusion.

Dr. He is a IET Fellow. He was selected as the Chinese Academy of Engineering Academician in 2013.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:53:07 UTC from IEEE Xplore. Restrictions apply.